

FERNS AND RELATIVES

Most ferns can be recognized by their graceful fronds, which unroll as they grow. Together with horsetails and whisk ferns, they make up a diverse and ancient group of non-flowering plants that reproduce by means of spores. Ferns grow in a wide variety of habitats, although the majority thrive where there is moisture and shade.

DIVISION	POLYPODIOPHYTA
CLASSES	1
ORDERS	11
FAMILIES	40
SPECIES	About 12,000

DEBATE LIVING FOSSILS

Horsetails have been around for over 300 million years. One of the largest extinct genera, called *Calamites*, grew up to 20 m (65 ft) tall, and had massive ridged stems over 60 cm (23½ in) across. The 20 or so species of horsetails that survive today are much smaller plants, growing in water, woodland, and disturbed ground. They look very different from ferns, producing their spores in cone-like structures at the tips of some of their stems, but most botanists now agree that they should be classified as an order within the ferns.

Some ferns are small and easily overlooked, but the group also includes large, tree-like species that can be over 15 m (50 ft) high. Many are compact plants, with a single cluster of fronds, but others have creeping stems that sprout fronds as they spread. Bracken – one of the most widespread creeping species – is particularly vigorous. Over many years, it can produce clumps, or clones, over 800 m (½ mile) across. Ferns also include species that live in fresh water and a wide variety of epiphytic species, which live on other plants.

Fern fronds are often finely divided, and usually develop from tightly coiled buds, known as croziers or fiddleheads. Spores are formed on the underside of the fronds, in structures that look like raised dots or lines. In many fern species, all the fronds produce spores, but some have two kinds of fronds: fertile fronds form spores, while sterile fronds are primarily adapted to collecting sunlight so the fern can grow. When a fern spore germinates, it develops into an intermediate stage known as a gametophyte.

Tiny, flat, and paper-thin, this eventually produces a new sporophyte – the spore-forming plant we recognize as a fern.

FERN RELATIVES

Fern taxonomy is in a state of flux, but all the scientific evidence suggests that ferns are more closely related to seed plants (gymnosperms and angiosperms) than to the lycophytes with which they were once grouped.

The fern division also includes horsetails, which are upright plants with hollow, cylindrical stems, surrounded by whorls of slender branches. Silica granules give them a rough texture, and they were once used for scouring pots and pans. Adder's tongues and whisk ferns are distinctive groups, with twig-shaped stems or a single, divided frond, that are also classed within the ferns.

✓ CHARACTERISTIC COIL

The delicate tips of young fern fronds are protected inside tight coils. These unwind as they stretch upwards towards the light.

